

Mammoth Tuff 67.5mi – Custom Plan for Monika Renk (6 weeks)

MAMMOTH TUFF

67.5 MILES

7500 FT

6-WEEK PLAN

MAMMOTH LAKES, CALIFORNIA

1 • Training Plan Brief

Welcome to your **Mammoth Tuff 67.5mi** training plan. This guide is built entirely from your questionnaire responses. Every number, every schedule, every recommendation is calibrated to your specific situation.

YOUR PROFILE

45

AGE

130 lbs

WEIGHT (59.0 KG)

5'1"

HEIGHT

7-10h

WEEKLY HOURS

10

YEARS RIDING

TBD

FTP (TEST IN WEEK 1)

YOUR WEEK, AT A GLANCE

Off days: Saturday — **Long rides:** Sunday — **Intervals:** Wednesday, Thursday

The day-by-day schedule lives in your training calendar — that is the source of truth.

WHAT MAKES THIS PLAN DIFFERENT

YOUR TRAINING METHODOLOGY: TRADITIONAL (PYRAMIDAL)

Your Traditional (Pyramidal) plan is built around a broad aerobic base with sustained climbing work through the middle and short VO2 efforts to raise the ceiling. By volume most of your riding stays genuinely easy (roughly 75%) — that's what builds the engine; the hard days are where this plan does its specific work. Matched to your hours and experience.

WHY THIS METHODOLOGY WAS SELECTED

- **7-10 hours/week** matches the Traditional (Pyramidal) approach
- **10 years** of cycling experience at **Intermediate** level
- **Traditional (Pyramidal)** training distribution (based on your available hours)

PROGRESSION STYLE

Progression: increase duration then density.

HEALTH NOTES

Sleep: good — **Stress:** moderate

PERFORMANCE EXPECTATIONS

With 7-10 hours per week over 6 weeks, you're building race-specific fitness using the Traditional (Pyramidal) approach. This plan is calibrated to your available time and experience level. Execute consistently, fuel properly, and trust the process.

YOUR GOALS & BLINDSPOTS

From your questionnaire — this is what we're building toward and what we're watching out for.

Success looks like: compete

2 • Training Zones

THE POINT OF ZONES

Training zones aren't arbitrary numbers. Each zone targets a specific physiological system. Training in the wrong zone doesn't just reduce effectiveness — it actively harms your progression by creating the wrong type of fatigue without the right type of adaptation.

Your plan controls how much time you spend in each zone — the zones are the vocabulary, the plan is the sentence. Your only job is to hit the prescribed zone when a workout calls for it. One caution that holds for every methodology: Zone 3 is the "gray zone" — too hard to recover from, too easy to create real adaptation. Ride it only when prescribed.

BEFORE YOU START: FTP TEST REQUIRED

Complete an FTP test before the structured intervals begin. Without your FTP, every zone target in this plan is a percentage of an unknown number. Until you test, use RPE (Rate of Perceived Exertion) to guide intensity.

After testing, recalculate all zones: **Zone watts = FTP × zone percentage.** If you use Zwift, TrainerRoad, or a Garmin — update your FTP setting immediately after testing.

Your threshold HR (LTHR): 165 bpm • measured HRmax 185 bpm. The HR zones below are a percentage of your LTHR — Zone 4 (Threshold) sits right around 165 bpm. Use power as the primary target and HR as the cross-check; on a hot or fatigued day, trust HR.

THE ZONE CHART

ZONE	NAME	POWER	% FTP	% LTHR	HR (BPM)	RPE	FEEL
1	Active Recovery	—	< 55% FTP	< 68% LTHR	< 112 bpm	1-2	Very easy, conversational. You should feel like you're barely working.
2	Endurance	—	56-75% FTP	69-83% LTHR	114-137 bpm	3-4	Easy effort. You can speak in full sentences. The bulk of your riding lives here.
3	Tempo	—	76-87% FTP	84-94% LTHR	139-155 bpm	5-6	Moderate. You can speak in short phrases. Comfortably hard.
GS	G Spot	—	88-93% FTP	92-96% LTHR	152-158 bpm	6-7	The G Spot between tempo and threshold. Maximum training stimulus with manageable fatigue.
4	Threshold	—	94-105% FTP	95-105% LTHR	157-173 bpm	7-8	Hard. Few words only. This is your FTP — sustainable for about 1 hour all-out.
5	VO2max	—	106-120% FTP	> 106% LTHR	> 175 bpm	9	Very hard, can barely speak. 3-8 minute efforts.
6	Anaerobic	—	> 120% FTP	N/A LTHR	—	10	Maximum effort. 30 seconds to 2 minutes.

FTP TESTING PROTOCOL

When an FTP test appears on your calendar, here's how to execute the 20-minute protocol. There is **one** maximal effort — the 20-minute test. Everything before it is preparation.

1. 12 minutes progressive warmup (Z1 building to Z2)
2. 3 minutes at RPE 7 (moderate-hard), then 1 minute easy — just opening the legs
3. **5-minute hard opener** — ride this hard (RPE 8-9), NOT all-out. Its only job is to clear out your anaerobic system so the 20-minute number reflects sustainable power, not a sprint. This is not the test.
4. 10 minutes easy recovery (Z1) — fully recover before the test

5. **20-MINUTE TIME TRIAL** — this is the test. Start at an effort you're confident you can hold, then build. Even pacing beats going out hard and fading. This sets your FTP.
6. 10 minutes easy cooldown

Formula: $\text{FTP} = \text{20-minute average power} \times 0.95$

WARNING

Don't test when tired. Do FTP tests when fresh — not after a hard training week. The test result sets ALL your training zones for the next 6 weeks. An inaccurate test means 6 weeks of wrong-zone training.

CRITICAL NOTES ON USING ZONES

- **Easy means EASY.** Zone 2 should feel almost too easy. If you're questioning whether you're going hard enough, you're in the right zone.
- **Hard means HARD.** Zone 4+ intervals should leave you breathless. If you can hold a conversation, you're not in the right zone.
- **Don't fill the gap.** The biggest mistake is spending too much time in Zone 3 (tempo). It feels productive but creates maximum fatigue with suboptimal adaptation.
- **Power > HR > RPE.** Use power meter if available, heart rate as backup, RPE as last resort. Heart rate lags — use RPE for short intervals.

3 • How Adaptation Works

Every workout in this plan exists for a reason. Understanding the adaptation model helps you make smart decisions when life disrupts the plan.

THE FOUNDATIONAL MODEL

STEP 1: STRESS

You apply a training stimulus (the workout). This creates controlled damage — micro-tears in muscle, glycogen depletion, cardiovascular stress.

STEP 2: RECOVERY

Your body repairs the damage. This takes 24-72 hours depending on workout intensity. Sleep, nutrition, and stress levels all affect recovery speed.

STEP 3: SUPERCOMPENSATION

Your body doesn't just repair — it overbuilds. You come back slightly stronger than before. This is the entire point of training.

STEP 4: REPEAT

Apply the next stress at the peak of supercompensation. Too soon = overtraining. Too late = detraining. The plan's timing handles this for you.

WHERE IT GOES WRONG

THE #1 TRAINING MISTAKE

Going too hard on easy days and too easy on hard days. This puts you in the "gray zone" — maximum fatigue, minimum adaptation. Trust the zones. Easy days should feel embarrassingly easy. Hard days should be genuinely hard.

THE #2 TRAINING MISTAKE

Skipping recovery. Adaptation happens during rest, not during workouts. The workout is the stimulus. Sleep is where you get faster. Cutting sleep to fit in more training is counterproductive — you're removing the adaptation window.

PRACTICAL RULES

- **Never add intensity to an easy day.** If the plan says Z2, stay in Z2. Even if you feel great.
- **Never add volume to a hard day.** Hard days are about quality, not quantity. If you can do more, you didn't go hard enough.
- **If you miss a workout, skip it.** Don't try to make it up by doubling the next day. The plan accounts for progressive overload — doubling creates injury risk.
- **If you're sick, stop.** Training while sick extends illness duration by 2-3x. Take the days off. You'll lose less fitness from rest than from a week of half-effort sick training.
- **Sleep > training.** If you have to choose between a 5am workout and 8 hours of sleep, choose sleep every time.

4 • Weekly Structure

Every week of your plan is built from the same four ingredients: a long ride, interval sessions, easy rides, and strength work. Your plan places them on specific days — follow the calendar. What matters here is understanding what each session type is *for*, because riders who understand the purpose execute better than riders who just follow orders.

Your week has 6 training days, 3 of which are key sessions. Key sessions are the ones that drive adaptation. Everything else supports them.

SESSION TYPES EXPLAINED

LONG RIDE

The backbone of endurance preparation. Primarily Zone 2 with race-specific efforts mixed in during Build and Peak phases. Build duration progressively — don't jump to your peak duration of 3.4-5.8 hours in Week 1.

INTERVALS

High-intensity sessions targeting specific energy systems. Follow the prescribed zones exactly. Full recovery between intervals is mandatory — cutting rest reduces the training stimulus.

EASY RIDE

Zone 1-2 only. Active recovery. The purpose is blood flow, not fitness. If you catch yourself hammering a Strava segment on an easy day, you're doing it wrong.

STRENGTH

Cycling-specific strength work. Focus on single-leg exercises, core stability, and hip strength. Keep it under 60 minutes. Heavy enough to be challenging, light enough to not wreck your legs for the next bike session.

WHEN LIFE INTERFERES

The plan owns the schedule. When real life collides with it, these principles keep the damage contained:

- **Protect the key sessions first.** If a week falls apart, keep one interval session and the long ride; let everything else go easy or disappear.
- **Never stack key sessions back-to-back** to "catch up" — that creates fatigue without proportional adaptation. A missed session stays missed.
- **Illness: drop ALL intensity.** Easy riding only if you feel up to it. Symptoms below the neck (chest, body aches) mean complete rest.

5 • Phase Progression

Your 6-week plan moves through 4 phases, each with a specific purpose. The exact week ranges are in your training calendar — what matters here is why each phase exists, so you don't rush base to get to the "hard stuff." Base fitness determines your ceiling.

BASE

The foundation. Mostly easy aerobic riding with progressive volume, plus strength work. This phase feels too easy — that's the point. Aerobic base determines your ceiling; there is no shortcut around it. Long rides build progressively. Your weekly budget limits ride length, so each long ride counts double — use them for race-pace fueling practice.

BUILD

Race-specific intensity enters: threshold and VO2max intervals, long rides with race-pace efforts. This is where it gets hard. Fatigue is expected — it's being managed deliberately through the recovery weeks your plan schedules.

PEAK

Maximum specificity. Race-simulation rides, dress-rehearsal fueling, intensity stays high while volume starts to drop. You should feel fitness arriving.

TAPER

Volume drops sharply; short, sharp efforts keep the engine awake. You cannot gain fitness now — you can only show up fresh or show up tired. Trust the training and rest.

RECOVERY WEEKS

Your plan schedules regular recovery weeks — volume drops 30-40%, intensity drops to easy riding only. These weeks are where adaptation actually happens. Skipping them is the fastest path to overtraining and stalled progress. They are not optional.

6 • Workout Execution

UNTIL YOU COMPLETE YOUR FTP TEST

Your ZWO files use FTP-based power targets. Without your FTP number, use RPE (Rate of Perceived Exertion) from the zone chart instead. After your Week 1 FTP test, update your FTP in Zwift/TrainerRoad/Garmin and all workout targets will calibrate automatically.

THE EXECUTION GAP

The plan says: "4x4min at 110% FTP with 4min recovery."

What actually happens:

- First interval: nailed it, felt strong
- Second interval: still good, maybe went a bit hard
- Third interval: struggling, cut 30 seconds short
- Fourth interval: abandoned or at 95% FTP instead of 110%

Result: You did a Zone 3-4 workout instead of a Zone 5 workout. Different stimulus, different adaptation, wrong training effect.

UNIVERSAL EXECUTION RULES

RULE 1: START CONSERVATIVE

First interval of every set should feel slightly too easy. If the plan says 110% FTP, start the first rep at 105%. Build into it. The goal is completing ALL intervals at target power, not crushing the first one and failing the rest.

RULE 2: QUALITY OVER QUANTITY

If you can't hold target power, STOP the workout. 3 good intervals at 110% FTP are worth more than 5 intervals where the last two are at 95%. Bad reps don't count — they just create fatigue.

RULE 3: RECOVERY MEANS RECOVERY

Between intervals, drop to Zone 1. Not Zone 2, not "easy Zone 3." Zone 1. Full recovery ensures the next interval hits the right energy system.

RULE 4: CADENCE MATTERS

Alternate between high cadence (95-105 rpm) and low cadence (60-75 rpm) intervals. High cadence trains the cardiovascular system; low cadence builds muscular force. A complete rider develops both — the range to spin smoothly when fresh and to push a big gear when the road tilts up.

INDOOR VS OUTDOOR WORKOUTS

Best done indoors: Short intervals (VO2max, anaerobic), FTP tests, cadence drills, recovery spins.

Best done outdoors: Long rides, tempo efforts, race simulations, skills practice, group rides.

Balance: Aim for at least 1-2 outdoor rides per week. Indoor training is efficient but doesn't prepare you for real-world race conditions — wind, terrain changes, group dynamics, and the mental challenge of a long day in the saddle.

WHEN AND HOW TO MODIFY WORKOUTS

- **Feeling great:** Don't add volume or intensity. Stick to the plan. Save the good legs for the next hard day.

- **Feeling tired:** Drop intensity by 5% but keep duration. If that's still too much, convert to Zone 2.
- **Feeling terrible:** Convert to easy spin or take the day off. One missed workout doesn't matter. Two weeks of subpar training does.
- **Missed workout rule:** Never try to make up a missed workout. Just move forward in the plan.

7 • Recovery Protocol

Adaptation happens during recovery, not during training. The workout is the stimulus. Recovery is where you actually get faster.

DAILY RECOVERY PROTOCOL

IMMEDIATELY POST-RIDE

- 24g protein + 59-71g carbs within 30 minutes (based on your 59kg body weight)
- Rehydrate: 150% of fluid lost (weigh before/after)
- Compression garments for rides > 3 hours

EVENING

- 8+ hours sleep (non-negotiable)
- No screens 60 min before bed
- Cool room (65-68°F / 18-20°C)
- Light stretching or foam rolling (10 min max)

AFTER HARD SESSIONS

- 10-minute cooldown spin (mandatory)
- Extra 200-300 calories in recovery meal
- No hard training for 48 hours
- Next day should be easy or rest

RECOVERY WEEKS

- Volume drops 30-40%
- No Zone 4+ intensity
- Extra sleep if possible
- This is where fitness consolidates

HRV: WHAT IT IS AND HOW TO USE IT

Heart Rate Variability (HRV) measures the time between heartbeats. Higher variability = better recovery. Lower variability = accumulated stress.

- **Good for:** Tracking long-term recovery trends (7-day rolling average), identifying overtraining before it happens
- **Not good for:** Making single-day training decisions, replacing how you feel
- **How to use:** Measure every morning before getting out of bed. If your 7-day average drops more than 10% below baseline, take an extra recovery day.
- **Apps:** HRV4Training, Elite HRV, Whoop

SIGNS YOU NEED MORE RECOVERY

- Resting heart rate elevated 5+ bpm above normal
- Can't hit power targets despite feeling "fine"
- Motivation drops for more than 3 consecutive days
- Sleep quality declining despite good sleep hygiene
- Persistent muscle soreness beyond 48 hours

- Getting sick more than usual

8 • Equipment Checklist

TRAINING EQUIPMENT

MANDATORY

- **Bike** — gravel or similar, in good working order
- **Helmet** — every outdoor ride, no exceptions
- **Water bottles / hydration pack** — two bottles for anything over 90 minutes outdoors; one within reach on the trainer
- **Repair kit (outdoor rides)** — spare tube, CO2, tire lever, multi-tool

RECOMMENDED

- **Power meter** — the single best training tool you can buy
- **Heart rate monitor** — chest strap preferred (wrist is less accurate)
- **Cycling computer** — for tracking intervals and zones in real-time
- **Foam roller** — for daily recovery

RACE DAY EQUIPMENT

- **Tires:** Test your race tires at least 3 weeks before race day. No new equipment on race day.
- **Nutrition:** Same bottles, same food, same pockets as your dress rehearsal ride
- **Clothing:** Weather-appropriate layers tested in training
- **Spares:** Two spare tubes, tire plugs, CO2 cartridges (minimum 2), multi-tool
- **Lights:** If there's any chance of finishing after dark

NOTHING NEW ON RACE DAY

No new shoes, no new saddle, no new nutrition, no new tires. Everything you use on race day must have been tested in training. This includes your exact nutrition plan — same brands, same quantities, same timing.

9 • Nutrition Strategy

YOUR PERSONALIZED TARGETS

65g/hr

CARB TARGET

330g

TOTAL RACE CARBS

Race-day range: 60-70g/hr

Hydration: 500-750 as a starting estimate, adjusted for heat and sweat rate - drink to thirst, and finish a shade lighter than you started, never heavier
ml/hr with 500-1000mg sodium per hour depending on sweat rate

Pre-race meal: 3-4 hours before start, high carb, moderate protein, low fat/fiber

You can have perfect training, a dialed bike, and excellent pacing strategy. None of it matters if you run out of fuel halfway through your race.

Fueling errors are the most common preventable cause of bad race days — the difference between finishing strong and crawling to the line.

THE REALITY CHECK

The cycling nutrition industry wants you to believe you need seventeen different products, each with proprietary blend ratios and specific timing windows measured in seconds. You don't.

You need carbohydrates during exercise. You need protein and carbs after exercise. You need reasonable daily nutrition that supports training. That's 95% of it. Everything else is optimization

you can worry about after you've nailed the basics.

DAILY NUTRITION FOR TRAINING

Your body is either recovering from the last workout or preparing for the next one. Daily nutrition supports both.

TIMING THAT ACTUALLY MATTERS

PRE-WORKOUT (2-3 HOURS BEFORE)

If training hard (threshold, VO2max):

- 1-2g carbs per kg bodyweight
- Low fiber, low fat, moderate protein
- Examples: oatmeal with banana and honey, or toast with peanut butter

If training easy (Z2 endurance):

- Eat normally, don't stress timing
- Can even train fasted if under 90 minutes

The rule: Hard sessions need fuel. Easy sessions are flexible.

POST-WORKOUT (0-90 MINUTES AFTER)

If workout was long (2.5+ hours) AND hard, AND you have another hard session within 24-36 hours:

- 20-30g protein
- 1-1.5g carbs per kg bodyweight
- Liquid is fine (protein shake, chocolate milk)

If workout was easy, short, or your next hard session is 48+ hours away:

- Just eat your next meal normally
- Recovery nutrition is optional

The rule: The more frequently you train hard, the more critical recovery nutrition becomes.

SUPPLEMENTS

Most supplements are placebo at best, actively harmful at worst.

WORTH TAKING

- **Vitamin D** (2000-4000 IU daily): Supports bone health, immune function, recovery. Most people are deficient. Get levels tested.
- **Creatine monohydrate** (5g daily): Improves high-intensity repeatability. Useful for VO2max and sprint work. Cheap, well-researched, safe.
- **Caffeine** (3-6mg/kg before hard sessions): Proven performance enhancer. Coffee works, pills work, gels work. Tolerance builds — cycle off occasionally.

NOT WORTH TAKING

- **BCAAs:** Complete waste if you eat adequate protein
- **Testosterone boosters:** Scams
- **Fat burners:** Scams with side effects
- **Recovery drinks with proprietary blends:** Overpriced protein + carbs

The rule: If you're deficient in something (Vitamin D, iron), fix it. If you're considering a supplement to “optimize,” ask yourself if you've already nailed sleep, nutrition basics, and training consistency. If not, fix those first.

FUELING DURING WORKOUTS

This is where races are won or lost.

For any ride over 90 minutes at moderate-to-high intensity (Z3+), you need 60-80g of carbohydrates per hour. Your gut can absorb approximately 60g of glucose per hour through SGLT1 transporters. Add fructose (which uses different transporters) and you can push to 90g total. The target for most athletes is 70-75g per hour — enough to fuel hard efforts without GI distress.

WORKOUT-SPECIFIC FUELING

WORKOUT TYPE	DURATION	CARBS/HOUR	NOTES
Z2 Endurance	2-4 hours	40-60g	Low intensity burns fat. Real food works great – PB&J, bananas, bars.
Tempo / G Spot	2-3 hours	60-80g	Start fueling at 30-45 min, not 90. Mix liquids and solids.
Threshold / VO2max	60-90 min	Pre-meal sufficient	Not depleting glycogen in 60 min. Maybe one gel mid-session.
Race Simulation	4-6 hours	70-80g	Practice your exact race-day fueling. Test products, timing, combinations.

The rule: The longer and harder the ride, the more critical fueling becomes. Easy rides are forgiving. Race-pace efforts are not.

DURATION-SCALED RACE FUELING

This is the key insight most nutrition advice gets wrong: **carb intake should scale DOWN with race duration, not up**. At lower intensities (longer races), your body shifts toward fat oxidation. Forcing 90g/hr of carbs into a system running at 44% VO2max exceeds physiological absorption capacity and causes GI distress.

RACE DURATION	CARBS/HOUR	FLUID/HOUR	PHYSIOLOGY
2-4 hours	80-100g	600-800ml	High intensity, carbs are dominant fuel
4-8 hours	60-90g	700-900ml	Classic endurance range (Jeukendrup 2014); racing hard with a trained gut justifies the top end
8-12 hours	50-70g	700-900ml	Fat oxidation increasing, GI risk climbing
12-16 hours	40-60g	750-1000ml	Reverse crossover point – fat is primary fuel
16+ hours	30-50g	750-1000ml	GI tolerance is the limiter, not energy

HYDRATION

You lose 0.5-2 liters of fluid per hour depending on temperature, humidity, and effort. Dehydration of 2-3% bodyweight impairs performance — for you at 130 lbs, that is 1.2-1.8 liters.

- **Aim for:** 500-750ml per hour (16-25 oz). More in heat, less in cold.
- **Short rides (<90 min):** Water is fine
- **Long rides (2+ hours):** Electrolyte drink with sodium (500-1000mg sodium/hour through sweat)
- **Don't overthink** potassium, magnesium, or trace minerals. Sodium is 90% of what matters for performance.

CRAMPING

Cramps are NOT caused by electrolyte deficiency (despite what the supplement industry tells you). Cramps are caused by neuromuscular fatigue — your muscles are tired and misfiring. Salt tabs might help by changing neuromuscular excitability, but the real fix is better pacing and better training.

TRAINING YOUR GUT

Your gut is trainable just like your muscles. If you never eat during training rides, your gut won't tolerate eating during races. SGLT1 transporters double in ~2 weeks of training (GSSI), but full adaptation takes 6-12 weeks.

- **Base phase:** Practice eating real food on easy rides. 40-50g carbs/hour, mostly solid. Build tolerance gradually.
- **Build phase:** Increase to 60-70g carbs/hour. Mix liquids and solids. Practice eating at tempo pace.
- **Peak phase:** Hit your race-day target. Race nutrition only (gels, drink, bars you'll use on race day). Practice at race pace.
- **Race week:** Stick with what worked in training. No new products. Trust your gut (literally).

RACE-DAY NUTRITION EXECUTION

Everything you practiced in training gets executed under stress.

PRE-RACE MEAL (3-4 HOURS BEFORE START)

Goal: Top off glycogen. Don't experiment.

- 2-3g carbs per kg bodyweight
- Moderate protein, low fat, low fiber
- Familiar foods only
- Examples: oatmeal with banana and honey, toast with jam and peanut butter, rice with eggs
- Drink 500ml water with your meal

STARTING LINE (30-60 MIN BEFORE START)

Take one gel (24g carbs) with 200ml water. You're not adding meaningful glycogen — you're making sure blood glucose is stable as the race starts.

THE FUELING TIMELINE

The most common mistake: waiting until you're hungry to start eating. By the time you feel hungry, your glycogen is already depleted and your brain is glucose-starved. You're in a hole you can't climb out of.

WINDOW	ACTION
0-30 min	Focus on pacing and positioning. Sip on drink, don't force nutrition yet.
30-60 min	First gel or equivalent (24g carbs). Start your fueling clock.
Every 30 min after	Consume carbs consistently. Gel every 30 min (48g/hr) + drink mix (20-30g/hr) = 70-80g total.
Aid stations	Top off bottles. Grab food if needed. Keep moving – you're not on a picnic.

Set a timer. Seriously. Your brain will be stupid. It will forget to eat. It will lie to you and say “I'm fine, I'll eat at the next aid station.” The timer removes decision-making.

WHEN YOUR STOMACH REBELS

It will. At some point in a long race, your gut will protest. High-intensity exercise diverts blood from your GI system to working muscles. Food sits there, undigested.

1. **Back off intensity for 5-10 minutes.** Drop to Z2 pace. Let gut blood flow recover.
2. **Switch to liquid calories temporarily.** Easier to digest than solids. Sports drink or cola at aid stations.
3. **Small sips, not big gulps.** Easier on the stomach, less sloshing.
4. **Don't panic and stop eating entirely.** You'll bonk 30 minutes later. Maintain some carb intake even if reduced.

What NOT to do: Hammer harder while nauseous. You'll either vomit or shut down completely.

WEIGHT MANAGEMENT VS PERFORMANCE

You want to be lean for racing. But chasing leanness during a training block is self-sabotage.

THE TRAINING BLOCK RULE

During your training build, your job is to train hard and recover well. Not cut weight. Energy deficit impairs recovery. Under-fueled training produces inferior workouts.

When to cut weight: After your race. During the off-season. When training volume is low and intensity is moderate. A 500-calorie daily deficit over 8-12 weeks can drop 4-6kg without destroying your fitness.

The rule: Chase performance metrics, not scale numbers. If your FTP and climbing times improve, your weight is fine. If they're declining, eat more.

THE RACE DAY FUELING RULE

On race day, eat what you've practiced. If you haven't tested a gel brand in training, don't eat it in the race. GI distress at mile 60 of a 67.5-mile race is a DNF-level problem that is 100% preventable.

10 • Mental Preparation

A 67.5-mile race is as much a mental challenge as a physical one. Your body will want to quit before it needs to. Your brain will lie to you about how bad it is. Having a mental strategy is not optional.

THE 6-2-7 BREATHING TECHNIQUE

When things get hard (and they will), use this breathing pattern:

6s

INHALE

2s

HOLD

7s

EXHALE

Deep belly breath through the nose. Brief pause at the top. Slow, controlled exhale through the mouth. Longer exhale = parasympathetic activation.

When to use: Before the start, during long climbs, when you want to quit, at aid stations, during the dark mile.

PROCESS GOALS VS OUTCOME GOALS

- **Bad goal:** "Finish in under 8 hours." (You can't control weather, mechanicals, or how you feel.)
- **Good goal:** "Fuel every 30 minutes. Stay in my zones. Ride my race." (100% within your control.)

On race day, focus exclusively on process goals. Check in every 30 minutes: Am I fueling? Am I hydrating? Am I in my zone? If yes, you're having a perfect race regardless of the clock.

MANTRAS THAT WORK

Pick 2-3 short phrases and practice them in training. Say them out loud on hard efforts.

- "This is what I trained for."
- "One pedal stroke at a time."
- "I've done harder things than this."
- "Fuel, hydrate, breathe. Repeat."

THE QUIT TEST

When you want to quit during the race (you will), ask yourself these three questions:

1. **Have I eaten in the last 30 minutes?** If no, eat first. You might just be bonking.
2. **Have I been drinking?** If no, drink 500ml. Dehydration mimics exhaustion.
3. **Am I actually injured or just uncomfortable?** Discomfort is part of the race. Actual injury (sharp pain, swelling) is a reason to stop.

If you've eaten, hydrated, and aren't injured — keep going. The feeling will pass.

11 • Race Week

Race week is about arriving at the start line fresh, confident, and prepared. You cannot gain fitness in the last week. You CAN lose it by doing too much. Your plan has the taper schedule — follow it. The principles below are what the schedule is protecting.

RACE WEEK PRINCIPLES

- **No fitness tests.** The urge to "check the legs" with one more hard effort spends race-day matches. Heavy legs mid-week are the taper working, not fitness leaving.
- **Nothing new.** No new equipment, food, or routines. Race week is execution, not experimentation.
- **Front-load the logistics.** Bike serviced, gear laid out, and bags packed early in the week — the night before is for sleeping, not wrenching.
- **Sleep is cumulative.** Bank 8-9 hours per night all week. The night before the race is usually bad for everyone — it also matters least if the week was good.

PRE-RACE CHECKLIST

- Bike serviced and in race-ready condition (do this by mid-week, not the night before)
- Tire pressure set for conditions (lower for gravel: 30-40 PSI depending on tire width)
- Nutrition packed and tested (same brands/products used in training)

- Bottles filled with your tested hydration mix
- Drop bags prepared (if applicable)
- Race number attached to bike or person
- Phone charged, emergency contacts accessible
- Weather checked — have a Plan B for layers
- Know aid station locations and what they provide
- Alarm set with buffer time (be at the start 45 minutes early)

CARB LOADING DONE RIGHT

Carb loading starts 2-3 days before the race. Increase carb intake to 8-10g/kg body weight. This isn't "eat everything in sight" — it's strategically increasing the carb portion of normal meals. Rice, pasta, bread, potatoes, fruit. Reduce fiber and fat intake to avoid GI issues.

12 • Race Day

RACE MORNING TIMELINE

TIME	ACTION
3 hours before	Wake up. Pre-race meal: 400-600 cal, low fiber/fat. Coffee if you normally drink it.
2 hours before	Sip 500ml water + electrolytes. Get dressed. Apply sunscreen and chamois cream.
1 hour before	Arrive at start. Check tire pressure. Warm up: 10-15 min easy spin with 2x30sec openers.
30 min before	Final bathroom stop. Eat a gel or banana. Get in position.
10 min before	6-2-7 breathing. Visualize your process goals. You're ready.
GO	Start conservative. First hour at 5% below target. Let others blow up – you'll pass them later.

RACE EXECUTION STRATEGY

- **First 25% of race:** Hold back. Ride 5% below target power/effort. Bank energy.
- **Middle 50% of race:** Settle into target effort. Fuel relentlessly. Every 20-30 minutes.
- **Final 25% of race:** If you feel good, increase effort slightly. If you're struggling, maintain. Do not blow up.
- **Fueling alarm:** Set a recurring timer for every 20 minutes. Eat whether you feel like it or not.

THE GOLDEN RULE

Start slower than you think you should. Every experienced endurance racer will tell you the same thing: the race doesn't start until the last third. Everyone goes out too hard. Be the one who doesn't.

POST-RACE

- Drink 1L fluid within 30 minutes of finishing
- Eat a real meal within 60 minutes (protein + carbs + fat)
- Light walking to keep blood flowing
- No training for 3-5 days minimum
- Celebrate. You just raced Mammoth Tuff 67.5mi. That's not nothing.

13 • Gravel Skills

Gravel racing rewards skill as much as fitness. A technically proficient rider at 3.5 W/kg will beat a 4.0 W/kg rider who can't handle loose corners or descents. These skills are trainable — but only if you practice them deliberately, not just ride through them.

CORNERING ON LOOSE SURFACES

THE WEIGHT SHIFT

On loose gravel, your front tire has less grip than on pavement. To corner safely:

- Drop your outside foot to 6 o'clock position
- Load the outside pedal — it does 60-70% of the work. Hands stay quiet; the inside hand should feel almost empty
- Keep the bike more upright than you would on pavement
- Look where you want to go — not at the obstacle you want to avoid

SPEED MANAGEMENT

All braking happens BEFORE the corner, not in it. On gravel:

- Brake early and progressively — no grabbing
- Release brakes before the apex
- Use both brakes — roughly 70/30 front-to-rear on hardpack, 65/35 on loose gravel; only mud and deep sand approach 50/50
- If the rear slides, do NOT grab the front brake

DESCENDING ON GRAVEL

BODY POSITION

- Hands in the drops — maximum brake leverage and control
- Elbows and knees slightly bent — act as suspension
- Weight shifted back on steep descents (hips behind saddle)
- One finger on each brake lever — maintain modulation
- Eyes scanning 15-20 feet ahead, not right in front of the tire

LINE SELECTION

- Smoother surface = faster, even if it's a longer line
- Avoid loose marbles on hardpack — ride the packed line
- On washboard: either go fast enough to float or slow enough to absorb
- Ruts: commit to being in or out — straddling a rut is how you crash

CLIMBING ON GRAVEL

TRACTION MANAGEMENT

Loose climbs are where most riders lose time. The key is steady, smooth power:

- Stay seated as long as possible — standing reduces rear traction
- Smooth pedal stroke — no surges, no mashing
- Higher cadence (75-85 rpm) reduces torque spikes that break traction
- Shift BEFORE you need to — shifting under load on gravel causes chain drops

WHEN TO STAND

- Only on hardpack or very short steep pitches
- When standing: keep weight forward, hands light on bars
- Shift to a harder gear before standing to maintain momentum
- If rear tire spins, sit immediately — you've lost traction

BIKE HANDLING DRILLS

Practice these weekly during easy rides. 10 minutes of drills per ride builds race-day confidence.

DRILL	HOW	WHY	TIME
Slow Speed Balance	Ride as slowly as possible without putting a foot down. Figure-8s in a parking lot.	Balance and core stability at low speeds – critical for technical sections.	3 min
One-Hand Riding	Ride with one hand on the bars, other on your thigh. Alternate every 30 seconds.	Eating, drinking, and adjusting kit while riding gravel requires single-hand confidence.	2 min each hand
Bunny Hops	Practice lifting both wheels over a stick or line on the ground. Don't need height – just timing.	Clearing obstacles (sticks, rocks, cattle guards) without stopping or swerving.	3 min
Emergency Stop	Ride at moderate speed, then stop as quickly as possible using both brakes. Practice on gravel and pavement.	Knowing your stopping distance on different surfaces. Muscle memory for panic stops.	2 min
Track Stand	Come to a stop and balance without putting a foot down. Start with slight uphill.	Group rides, aid stations, and intersections without unclipping.	2 min

TIRE PRESSURE FOR GRAVEL

Tire pressure is the single biggest performance variable on gravel – more impactful than any component upgrade you can buy.

GENERAL GUIDELINES

- **Tubeless:** Always run tubeless for gravel. Period.
- **38-42mm tires:** 28-35 PSI for most riders
- **45-50mm tires:** 25-30 PSI for most riders
- **Lighter riders (<150 lbs):** 2-4 PSI less than above
- **Heavier riders (>180 lbs):** 2-4 PSI more than above
- **Rear tire:** 2-3 PSI more than front (more weight)

CONDITIONS ADJUSTMENTS

- **Wet/muddy:** Drop 2-3 PSI for grip
- **Dry hardpack:** Can run 2-3 PSI higher for speed
- **Rocky/chunky:** Drop 2-3 PSI to absorb impacts and prevent pinch flats
- **Mixed surface:** Optimize for the worst surface you'll encounter

TEST YOUR PRESSURE BEFORE RACE DAY

Find your sweet spot by testing different pressures during long training rides. Drop 2 PSI at a time and note how the bike feels. Too low = squirmy in corners, too high = harsh and bouncy. The right pressure feels fast AND confident. Write it down. Use that pressure on race day.

MECHANICAL SKILLS

You must be self-sufficient on gravel. There are no team cars. Practice these before race day:

- **Flat repair:** Fix a flat (tube or tubeless) in under 5 minutes. Practice at home.
- **Chain repair:** Know how to use a chain tool. Carry a quick link.
- **Derailleur adjustment:** Know how to adjust limit screws if your derailleur gets hit.
- **Tire plug:** Practice plugging a tubeless tire with a plug kit. It takes 30 seconds once you've done it.

THE 5-MINUTE RULE

If you can't fix it in 5 minutes, you can't fix it on the course. The skills above cover 95% of mechanicals. The other 5% (cracked frame, destroyed wheel) require a ride to the nearest aid station. Carry a phone and know the course map.

WHAT TO CARRY ON RACE DAY

Pack these the night before. Check them at the start line. No exceptions.

ON THE BIKE

- 2 water bottles (filled with electrolyte mix)
- Frame bag or saddle bag with repair kit
- Cycling computer with route loaded

REPAIR KIT

- Spare tube (correct size and valve type)
- Tubeless plug kit (Dynaplug or similar)
- 2x CO2 cartridges + inflator head
- Tire lever (just one is fine)
- Multi-tool with chain breaker
- Quick link for your chain

NUTRITION

- Gels/chews for first 2 hours (tested in training)
- Solid food for hours 3+ (rice cakes, bars)
- Electrolyte tabs for refills at aid stations
- Emergency caffeine gel for the final push

SAFETY

- Fully charged phone in waterproof case
- Emergency contact card in jersey pocket
- Cash or card (some aid stations are gas stations)
- Sunscreen applied before the start

14 • Heat Training

Mammoth Tuff's typical conditions run cool, so this isn't survival prep — but our coaching read is the block is still worth it: plasma volume and aerobic headroom carry over to any race-day weather.

HEAT TRAINING: RECOMMENDED

Heat acclimation reliably helps performance IN THE HEAT. The cool-condition evidence is genuinely split in the research — some studies show a small aerobic gain (plasma volume expansion, hemoglobin mass), others show none — which is why we call this recommended, not required. Skip it only if recovery is already stretched thin.

THE PROTOCOL

HEAT PROTOCOL (WEEKS 3-4 OF YOUR PLAN)

A concentrated 2-week block, ending close to race week so the adaptations don't fade before you need them — they decay at roughly 2.5% per day once you stop.

- **5-6 sessions across the 2 weeks** — roughly every other day, not daily.
- **Post-exercise passive heat:** 15-20 min sauna, hot bath, or extra layers during/after a normal training session. Add duration gradually.
- **What adapts first:** plasma volume expansion and earlier sweat onset show up in the first few sessions; sweat-rate and core-temperature adaptations mature by the end of the block.

HEAT SAFETY

- Weigh yourself before and after — replace fluid losses, don't guess
- Never stack a heat session on a hard training day or when you're sick
- Stop immediately for dizziness, nausea, confusion, or goosebumps in the heat
- Get medical clearance first if you're on cardiovascular medication or pregnant

15 • Altitude Training

Mammoth Tuff takes place at significant altitude. The start elevation is around 8100 feet, with racing at 8100+ feet. This has real, measurable effects on performance that you cannot ignore.

THE ALTITUDE REALITY

At 8100 feet, expect approximately **5.1% reduction in FTP** compared to sea level. Your heart rate will run higher at the same power output. RPE becomes unreliable — use a power meter. Recovery between efforts takes longer. Sleep quality may suffer for the first 2-3 nights.

THE PHYSIOLOGY

WHAT ALTITUDE DOES TO YOUR BODY

- **Reduced oxygen availability:** At 7,000 ft, there's ~22% less oxygen per breath compared to sea level
- **Higher heart rate:** Your heart pumps faster to compensate, running 5-15 bpm higher at same effort
- **Faster lactate accumulation:** You reach threshold sooner with less absolute power
- **Impaired recovery:** Between intervals, between efforts, between days — everything takes longer
- **Dehydration risk:** Dry mountain air + increased respiration rate = faster fluid loss
- **Increased calorie burn:** Your body works harder at altitude, burning 5-10% more calories

POWER LOSS BY ELEVATION

ELEVATION	FTP REDUCTION	IMPACT
Sea level - 3,000 ft	0%	No significant effect
3,000 - 5,000 ft	1-2%	Barely noticeable for most athletes
5,000 - 7,000 ft	2-4%	Noticeable on hard efforts
7,000 - 9,000 ft	4-6%	Significant – requires pacing adjustment
9,000 - 11,000 ft	6-8%	Major impact – acclimatization critical
11,000+ ft	8-12%+	Severe – multi-week acclimatization needed

ACCLIMATIZATION PROTOCOL

IF YOU CAN ARRIVE EARLY (IDEAL: 10-14 DAYS BEFORE)

- Days 1-3: Easy activity only. Walk, light spin. Let your body adjust to breathing.
- Days 4-7: Light training at reduced intensity (80% of normal power targets).
- Days 8-14: Gradually return to normal training intensity. You'll feel closer to normal.
- Key: Hydrate aggressively (3-4L/day), sleep extra, avoid alcohol.

IF YOU CAN'T ARRIVE EARLY (THE REALISTIC OPTION)

Most athletes can't take 2 weeks off before a race. Here's how to mitigate:

- **Arrive as late as possible** — 24-48 hours before is better than 3-5 days. The worst window is 2-5 days at altitude (past the adrenaline boost, before any adaptation).
- **Heat training offers partial crossover:** see the Heat Training section — some of its adaptations (plasma volume expansion, earlier sweat onset) partially transfer to altitude too.
- **Pace conservatively:** Start 10% below target power. Altitude punishes going too hard early more than sea-level racing does.
- **Hydrate aggressively:** Start hydrating 48 hours before the race. Drink 500ml+ more than normal daily.

HEAT CROSSOVER

Heat training isn't required prep for this race (see the Heat Training section for the honest framing on that), but if you do it anyway, some of the same adaptations — plasma volume expansion in particular — partially transfer to altitude too. The evidence for a benefit outside hot conditions is genuinely mixed, not settled.

RACE DAY AT ALTITUDE

ALTITUDE PACING RULES

- Start 10% below your sea-level target power
- Use power meter, not RPE — RPE is unreliable at altitude
- Recovery between hard efforts takes 20-30% longer
- Be judicious with matches early — altitude limits recovery
- Fuel and hydrate more aggressively than at sea level (5-10% more calories, 500ml+ more fluid)
- Expect your heart rate to be 5-15 bpm higher at the same power

16 • Women-Specific Considerations

If you're a woman training for gravel racing, your physiology is different from men's in ways that actually affect training and performance. This isn't patronizing — it's honest acknowledgment of real differences that matter.

The good news: these differences are trainable and manageable. The bad news: if you ignore them, you're making things harder than they need to be.

MENSTRUAL CYCLE & TRAINING

Your menstrual cycle affects training capacity. Not as an excuse to skip workouts, but as a variable to monitor and work with.

FOLLICULAR PHASE (DAYS 1-14, START OF PERIOD TO OVULATION)

What's happening: Estrogen rises, body temperature is lower, insulin sensitivity is higher.

Training impact: This is typically your power window. You'll recover faster, handle intensity better, and feel stronger. Days 5-14 are often your best training days of the month.

How to use it: Schedule your hardest interval sessions, longest rides, and FTP tests during this phase when possible. Your body is primed for high-quality work.

LUTEAL PHASE (DAYS 15-28, OVULATION TO NEXT PERIOD)

What's happening: Progesterone dominates, body temperature rises (~0.5°F), metabolism shifts toward fat burning, inflammation increases.

Training impact: Recovery takes longer. Interval quality might decline. You might feel flat even when doing everything right. Heart rate runs 5-10 bpm higher at same effort. Carb cravings increase (because your body actually needs more carbs — progesterone reduces glycogen storage efficiency).

How to use it: This is not the time to test FTP or push for breakthrough sessions. Focus on base miles, maintenance intervals, and recovery. Listen to your body more carefully. If a training week falls during late luteal phase and you feel like garbage despite perfect execution, that's normal — it's hormones, not fitness loss.

IRON CONSIDERATIONS

Monthly blood loss means monthly iron loss. Women who menstruate need roughly 18mg of iron daily (vs. 8mg for men). Athletes need even more due to foot-strike hemolysis, GI losses, and increased red blood cell turnover.

Low iron = compromised training: Fatigue, inability to hit power targets, poor recovery, elevated heart rate, shortness of breath during efforts you should handle easily.

- Get bloodwork annually (ferritin, serum iron, hemoglobin, hematocrit)
- Target ferritin >50 ng/mL for athletes (many docs say >15 is "normal" — that's too low for performance)
- Iron-rich foods: red meat, dark leafy greens, lentils, fortified cereals
- Consider supplementation if levels are low (but get tested first — too much iron is also a problem)
- Take iron with vitamin C (aids absorption), avoid taking with calcium (blocks absorption)

Your period is not an excuse to skip workouts, but it IS a variable to monitor. Track your cycle. Notice patterns. Adjust expectations during luteal phase. Capitalize on follicular phase. This is performance optimization, not weakness.

FUELING DIFFERENCES

Women's bodies process fuel differently than men's, especially during exercise.

Women need MORE carbs relative to body weight than men. Despite often being told to eat less, female athletes actually need aggressive carbohydrate intake to support training and maintain hormonal health.

Why: Women's bodies preferentially spare carbohydrate and burn more fat at rest. Sounds great, except during high-intensity efforts (most of gravel racing), you **NEED** carbs. If you're chronically under-fueling carbs, your body will:

- Downregulate thyroid function (slower metabolism, more fatigue)
- Disrupt menstrual cycle (late periods, missed periods, longer cycles)
- Compromise bone density
- Tank performance

Your training day target: 354-413g carbs (59.0kg x 6-7g/kg). More on long ride days: 472-590g (59.0kg x 8-10g/kg).

Racing target: 60-80g carbs per hour for rides over 90 minutes at moderate-to-high intensity (scaled down for longer durations — see Nutrition Strategy section). Don't under-fuel trying to "stay lean" — that strategy kills performance AND health.

Women thermoregulate differently than men. This matters for gravel events.

Women sweat less than men at the same relative intensity. Sounds like an advantage (less fluid loss), but it's not — it means core temperature rises faster because evaporative cooling is less efficient.

The result: Women reach critical core temperature thresholds earlier in hot conditions, leading to earlier performance decline, higher perceived exertion at same power, greater cardiovascular strain, and increased risk of heat illness.

- **Pre-cooling is critical:** Cold water immersion, ice vests, cold drinks before start
- **Aggressive cooling during race:** Ice in jersey pockets, cold water over head/neck at aid stations
- **Heat acclimatization training:** See the Heat Training section for the full protocol and whether it's required, recommended, or optional for your race
- **Monitor core temp signals:** Goosebumps, chills, confusion, nausea = pull back immediately

WHY WOMEN MAY NEED MORE RECOVERY

Women often need MORE recovery than men at the same relative training intensity. This isn't weakness — it's physiology.

- Estrogen has anti-inflammatory properties, but it also affects tissue repair timing
- Lower testosterone means slower muscle protein synthesis
- Higher Type I muscle fiber percentage means different recovery demands
- Hormonal fluctuations (especially luteal phase) increase baseline inflammation

What to do:

- Pay closer attention to RHR and HRV trends
- Don't try to match training partners if they're recovering faster — train YOUR body, not theirs
- Prioritize sleep (7-9 hours minimum, non-negotiable)
- Don't skip rest weeks
- Back-to-back hard days might wreck you more — you might need an extra rest day per week

IMPORTANT

This plan is NOT designed for pregnant or postpartum athletes.

If you're pregnant: Stop this plan immediately. Consult your physician before continuing ANY structured training. Racing is off the table, but moderate training may be safe with medical clearance.

Postpartum: Wait at least 6 months before attempting race-specific training. Get clearance from OB/GYN and ideally a pelvic floor physical therapist. Rebuild base fitness gradually — don't jump back into high-intensity work.

EQUIPMENT CONSIDERATIONS

- **Sports bra:** Non-negotiable. Test on long rides, not just in the store. If it's uncomfortable for 30 minutes, it'll be unbearable at mile 89. High-impact support is essential — you're bouncing on gravel for hours.
- **Chamois:** Women's anatomy requires women-specific design. This is not marketing — the padding placement and shape are different. A men's chamois will cause problems. Test multiple brands if needed. Numbness, chafing, or pain means wrong fit.
- **Saddle:** Proper bike fit and saddle selection are critical. Women's sit bone width is often (but not always) wider than men's. Get a professional fit if you're experiencing numbness, pain, or discomfort. These issues don't resolve with more miles — they require equipment changes.

THE BOTTOM LINE

Your body is different. Not worse, not weaker — different. Train accordingly.

- Track your cycle. Work with it, not against it.
- Fuel aggressively. Don't under-eat trying to stay lean.
- Take heat management seriously. You're at higher risk in hot conditions.
- Prioritize recovery. You might need more than male training partners.
- Get proper equipment. Generic gear won't cut it.

Racing gravel is hard for everyone. Being a woman doesn't make it harder — but ignoring female physiology makes it unnecessarily difficult. Do the work. Respect your body's signals. Show up prepared.

GRAVEL GOD

Custom training plans for gravel racing

6 Weeks • ENDURE Plan Engine